

BASIC NON-LIFE INSURANCE COURSE

Engineering Insurance

HISTORICAL BACKGROUND

Engineering Insurance had its beginning during the 18th century industrial revolution in Great Britain

The factory system of production became established in this century and accidents arising out of the use of steam plants were common

The first boiler insurance company, the Steam Boiler Assurance Company, was formed in 1858.

Importance of Construction Insurance

Offers comprehensive protection against loss to:

- Contract Works
- Construction Plant and Equipment
- Construction Machinery
- Liability to Third Party

Provides financial stability to all parties in the Construction contract.

Provide protection to investors.

Serves as collateral and recourse of Financial Institutions.

Classes of Engineering Insurance

Non-renewable
Classes

Renewable
Classes

Non-Renewable Classes

- **CONTRACTORS ALL RISKS (CAR)**
- **ERECTION ALL RISKS (EAR)**

Renewable Classes

- **MACHINERY BREAKDOWN (MB)**
- **BOILER & PRESSURE VESSEL
EXPLOSION**
- **ELECTRONIC EQUIPMENT**
- **INSURANCE (EEI)**
- **DETERIORATION OF STOCKS (DOS)**
- **CONSEQUENTIAL LOSS**

Classes of Engineering Insurance

Renewable Classes

These classes are intended to cover completed industrial plants and installations after their turnover and acceptance by the owner.

Classes of Engineering Insurance

Non - Renewable Classes

- designed to cover buildings and other civil engineering projects during construction as well as machinery and equipment during erection or installation.
- cover commences on delivery of the materials and machinery to the site of erection or installation.
- terminates when erection, installation, testing and commissioning are completed and the plant is ready for normal use by the purchaser.

C O N S T R U C T I O N C O N T R A C T

TERMS COMMONLY USED

- Principal
- Bar Chart
- S-Curve
- Policy Period
- Period of Cover
- Third Party
- Contract Price
- Average Clause
- Materials Supplied by the Principal
- Principal Controlled Project/s
- Vertical Projects
- Horizontal Projects
- Wet Risks
- Testing and Commissioning

RELEVANT INFORMATION IN THE CONSTRUCTION CONTRACT

- The name and the details of the Principal
- The names of all Contractors, Sub-Contractors and Specialty Contractors
- Nature of Works
- Location / Off-site Locations
- Duration of Construction
- Total Contract Price
- Insurance Requirements
- Bills of Materials/ Breakdown of Contract Value
- Bar Chart
- S - Curve

CONTRACTORS ALL RISK INSURANCE (CAR)

Contractors All Risk
Insurance (CAR)

CONTRACT WORKS

- **Permanent Works**
- **Temporary Works**
- **Materials Supplied by the
Principal**

CAR Insurance primarily covers Civil Works !

PROJECTS CONSIDERED AS CIVIL WORKS

- **Building Constructions (Offices, Hospitals, Schools, Theatres, etc.)**
- **Constructions of Roads and Bridges**
- **Railways, Ports and Harbors**
- **Airport Complex**
- **Dams, Canals, Irrigation Projects**
- **Residential Complex**
- **Commercial Complex**
- **Warehouse Complex**
- **Industrial Complex**

PERIOD OF COVER

For the CAR insurance, coverage starts upon the commencement of work or after the unloading of the items at the site. Coverage expires on the date specified in the schedule or when taken over or put into service whichever is earlier.

Contractors All Risk
Insurance (CAR)

PERIOD OF COVER

Start of Contract Works
(Unloading of Materials)

Completion of Works



ERECTION ALL RISK INSURANCE (EAR)

Erection All Risk Insurance (EAR)

PROJECTS CONSIDERED AS ELECTRO - MECHANICAL WORKS

Installations of :

- **Petrochemical Plants**
- **Power Generators**
- **Manufacturing Plants**
- **Steam Boilers**
- **Turbines**
- **Electrical Transmission Lines**
- **Conveyor Systems**
- **Power Machineries**
- **Control System**
- **Pumps, Motors, Transformers**

CONTRACT WORKS UNDER EAR POLICY

- **Permanent Erection Works**
 - **Freight**
 - **Customs Duties & Dues**
 - **Cost of Erection or Installation**
- **Temporary Works**
- **Civil Engineering Works**
- **Material Supplied by the principal**

Erection All Risk Insurance (EAR)

- ERECTION OF POWER PLANTS
- TRANSFORMERS
- COMPRESSORS
- SMELTING PLANTS
- CONVEYOR SYSTEMS
- ELEVATORS AND ESCALATORS
- AIR CONDITIONING SYSTEMS



PERIOD OF COVER

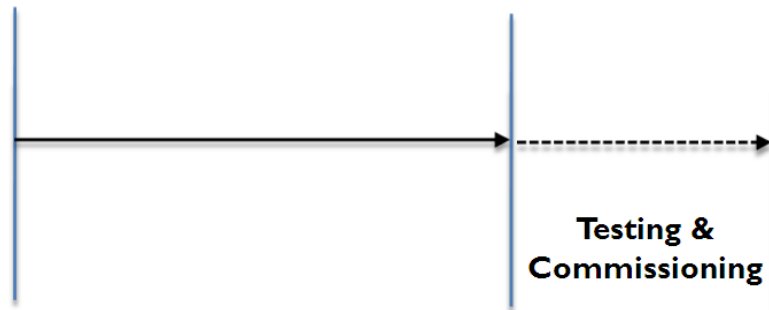
For the EAR insurance, coverage starts upon the commencement of work, or after the unloading of the items at the site. Coverage expires on the date specified in the schedule, or immediately after the test loading is completed, or when taken over or put into operation whichever is earlier.

Erection All Risk Insurance (EAR)

PERIOD OF COVER

Start of Contract Works
(Unloading of Materials)

Completion of
Works



SUM INSURED

The sum insured for material damage is the total project cost, or each item at the completion of the construction or erection inclusive of freight, custom duties, erection costs and materials supplied by the principal

COMPARATIVE ANALYSIS CAR & EAR INSURANCE

SUBJECT MATTER:

- **CAR INSURANCE**
 - Covers primarily Civil Works
 - Electro-mechanical components by endorsements to cover testing and commissioning (optional)
- **EAR INSURANCE**
 - Covers primarily Electro-mechanical works
 - Automatically covers Civil Work components of the Installation project
 - Provides cover for Testing and Commissioning for Brand New Machineries & Equipment to be installed

SCOPE OF COVERAGE

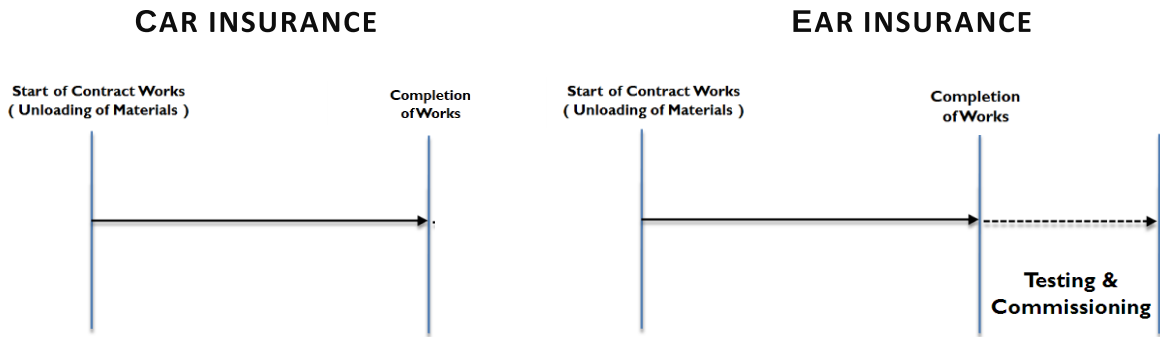
Section I - MATERIAL DAMAGE

Section II - THIRD PARTY LIABILITY

CAR AND EAR

COMPARATIVE ANALYSIS CAR & EAR INSURANCE

PERIOD OF COVER:



CAR AND EAR

INSURING PROVISIO

" THE POLICY WILL INDEMNIFY THE INSURED FOR
ALL SUDDEN AND UNFORESEEN LOSSES NOT
OTHERWISE EXCLUDED "

**Unforeseen and sudden physical loss of or damage to
insured items during the course of their construction,
from whatever cause not specifically excluded, are
covered by the policy**

CAR AND EAR

WHO CAN BE INSURED?

- Principal
- General Contractor
- Sub - Contractors or Specialty Contractors
- Financial Institutions
- Project Manager / Consultants
- Manufacturers and/or Suppliers
- etc.

CONSTRUCTION PLANT AND EQUIPMENT

- Accommodations
- Field Offices
- Storage Facilities
- Aggregate Crushers
- Cements, Asphalt Mixing Plant
- Utilities (Electricity, Water, etc.)
- Scaffoldings

CAR AND EAR

CONSTRUCTION MACHINERY

- Earth Moving Equipment
- Graders, Loaders, Excavators
- Cranes
- Pile Drivers
- Elevators, lifts
- Other vehicles not licensed for road use

SCOPE OF COVERAGE

SECTION I - MATERIAL DAMAGE

- Contract Works
 - Contract price
 - Materials Supplied by the Principal
- Construction Plant and Equipment
- Construction Machinery
- Storage Facilities
- Surrounding Properties of the Insured
- Properties in Care, Custody and Control of the Insured (*)
- Cost of Clearance Debris

CAR AND EAR

SECTION II - THIRD PARTY LIABILITY

- **Property Damage**
- **Bodily Injury**

CAR AND EAR

EXCLUSIONS

Loss or Damage due to :

- War, Rebellion, Insurrection, Invasion, Confiscation, Nationalization
- Consequential Loss, Penalties, Loss due to Delay, Loss of Contract
- Willful Act or Willful Negligence of the Insured
- Nuclear Reactions, Explosions, Radiation or Radioactive Contamination
- Faulty Design, Cost of Replacement of Defective Workmanship or Material other than faults in Erection
- Loss to Vehicles Licensed for General Road Use or Waterborne Vessels or Aircraft
- Loss to File, Drawings, Bills, Currencies, Deeds and Checks
- Loss discovered only at the time of taking of an Inventory
- Cessation of Work whether Total or Partial
- Deductible stated in the Schedule

It is however possible to delete various exclusions and provide insurance cover by special agreement

EXAMPLES OF PERILS COVERED UNDER CAR & EAR

- Fire, lightning and aircraft impact
- Accidental damage during assembly or construction due to faulty manipulation, dropping or falling, etc.
- Collapse, collision and impact
- Burglary, theft and malicious damage
- Water damage, flood, storm and tempest
- Electrical and mechanical breakdown and explosion
- Damage due to defects in material, casting, workmanship, design, plan and specification (but costs incurred in rectifying the defect would normally be excluded)
- Riot, strike, subsidence, landslide, other perils resulting from acts of nature

EXTENSIONS TO STANDARD COVER

- Cross Liability - Where the contract wordings stipulates that the insurance is to be for joint interests (principal, contractor and subcontractors) third party insurance can be granted to each party separately. This allows one insured party to take action against another and seek indemnity.
- Maintenance Period Cover - The insurer's liability during the maintenance period is limited to loss or damage caused by the insured in the course of operations, carried out for the purpose of complying with the obligations under the maintenance provisions of the contract.

EXTENSIONS TO STANDARD COVER

- **Professional Fees** - following a loss, repair or reinstatement of contract or erection works may necessitate the employment of professional services such as architects or consulting engineers.
- **Inland Transit** - loss or damage occurring during transit (not sea or air) to the contract site including loading and unloading and any transshipments storage or deviations en route.
- **Expediting Expense** - covers the extra cost of overtime rates of wages, express delivery or airfreight, in connection with repairs or replacements covered under the policy

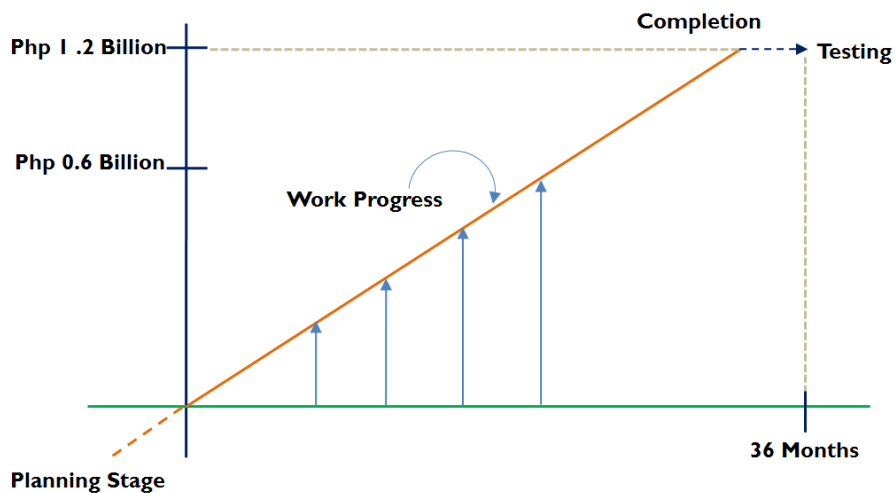
RISK ASSESSMENT/UNDERWRITING CAR AND EAR

- **Assess the contractor and subcontractors** in terms of their expertise or experience in the project to be constructed, the moral risk, and the loss experience
- **Review contract conditions** - determine the respective responsibilities of the principal and contractors, which perils must be insured, for what sum the insurance is to be provided, the duration of the project and the scope of maintenance period liability
- **Assess Work Schedule** and the method of execution of the work
- **Review Period of Construction** – Contractors prepare a plan that indicates the work progress against the total duration of the project. Usually this is a PERT (Program Evaluation and Review Technique) and CPM (Critical Path Method) or Bar Charts

RISK ASSESSMENT/UNDERWRITING CAR AND EAR

- Assess the Site. Factors to be considered here are the location, exposure to extraneous perils, geological characteristics and facilities for fire-fighting.
- Review the Plans. Shows the work to be done including design.

UNDERWRITING & RATING CONSIDERATIONS



CAR vs EAR

Both the CAR and EAR provide:

a comprehensive and adequate protection against loss or damage to contract/erection works, construction plant and equipment and/or machinery,

as well as for third party claims (property damage or bodily injury) arising in connection with the execution of a construction or erection work.

EAR also covers testing and commissioning of the machinery/equipment installed.

CAR and EAR are both types of “all risks” insurance.

ELECTRONIC EQUIPMENT INSURANCE (EEI)

Electronic Equipment Insurance is an “all risks” coverage for electronic equipment ; such as:

Electronic Data Processing Equipment (EDP Equipment)

Communication facilities such as telephone exchanges, radio, television and communication apparatus, etc.

Electrical and radiation equipment for medical use such as electrocardiograph, X-ray and other radiation equipment in hospitals and medical laboratories, etc.

Miscellaneous equipment such as film and television studio equipment, automatic addressing, booking and broadcasting, machines, etc.

**ELECTRONIC EQUIPMENT
INSURANCE (EEI)**

- EDP Equipment/Computers (Banks, Offices)
- Telephone Control and Relay Equipment
- BROADCAST EQUIPMENT
- Other Electronic/Computerized Equipment used in various fields of Business
- Medical Equipment

ELECTRONIC EQUIPMENT
INSURANCE (EEI)

SCOPE OF COVERAGE

Section I - MATERIAL DAMAGE

section II - EXTERNAL DATA MEDIA

Section III - INCREASED COST OF WORKING

ELECTRONIC EQUIPMENT
INSURANCE (EEI)

SCOPE OF COVERAGE

SECTION I - MATERIAL DAMAGE

- fire, lightning and explosion, falling aircraft
- smoke, soot, corrosive gases
- water and humidity
- short-circuit and other electrical causes
- design, manufacturing, assembly and erection faults
- defects in casting and material, workshop errors, bad workmanship
- faulty operation, lack of skill, gross negligence
- malicious acts of workmen, employees, third parties
- burglary and theft
- hail, frost, typhoon and storm
- earthquake, subsidence, landslide, rockslide, tsunami

ELECTRONIC EQUIPMENT
INSURANCE (EEI)

TYPICAL INSURED PERILS

- **Lightning Damage**
- **Power Surge Damage**

ELECTRONIC EQUIPMENT
INSURANCE (EEI)

SCOPE OF COVERAGE

Section I - MATERIAL DAMAGE

- **SUM INSURED** - The sum insured must always be the replacement value of the insured electronic equipment (value of the new item plus custom duties, transportation and installation charges, etc.
- **INDEMNIFICATION** - The Insurers shall indemnify the Insured by payment of the expenses required for the repair of the damage.

ELECTRONIC EQUIPMENT INSURANCE (EEI)

TYPICAL EXCLUSIONS

- **War Perils, Riot, Strike, Civil Commotion, etc.**
- **Nuclear Perils**
- **Willful Act or Willful Negligence of the insured or of his representatives**
- **Loss or Damage caused by any faults or defects existing at the time of commencement**
- **Wear and Tear, Oxidation, Rusting, Erosion, Corrosion,**
- **Theft**
- **Earthquake, Volcanic Eruption, tsunami, storm, typhoon, Cyclone, Hurricane**
- **Replacement of Expendable Parts**
- **Deductible stated in the schedule**

SCOPE OF COVERAGE

ELECTRONIC EQUIPMENT

Section II – EXTERNAL DATA MEDIA

INSURANCE (EEI)

- **SCOPE OF COVER** - Provides cover for both the material value of the external data media and the cost of restoration of lost information due to the perils under Section I.
- **SUM INSURED** - The sum insured under this section is the material value of the insured external data media plus the estimated restoration cost of lost information.
 - Value of External Media
 - Additional Manpower
 - Rental Cost of Alternative Premises
 - Overtime / Holiday Pay
 - Rental Equipment
 - Etc.
- **INDEMNIFICATION** - The Insurers shall indemnify the Insured for the cost of replacement of the damaged data media plus the actual cost of restoring lost information.

SCOPE OF COVERAGE

ELECTRONIC EQUIPMENT
INSURANCE (EEI)

Section III – INCREASED COST OF WORKING

- **SCOPE OF COVER** - Covers additional expenses that may arise if the Insured's computer breaks down and it becomes necessary to hire substitute computer facilities to maintain the normal processing activities during the interruption period.
- **SUM INSURED** - The sum insured must be equal to the cost of hiring a substitute computer for the duration of the interruption period.
- **INDEMNIFICATION** - The Insurers shall indemnify for the necessary expenditure for the use of a substitute computer equipment during the interruption period over and above the normal expenses in the operation of the system.

END OF SESSION
Thank you for listening.